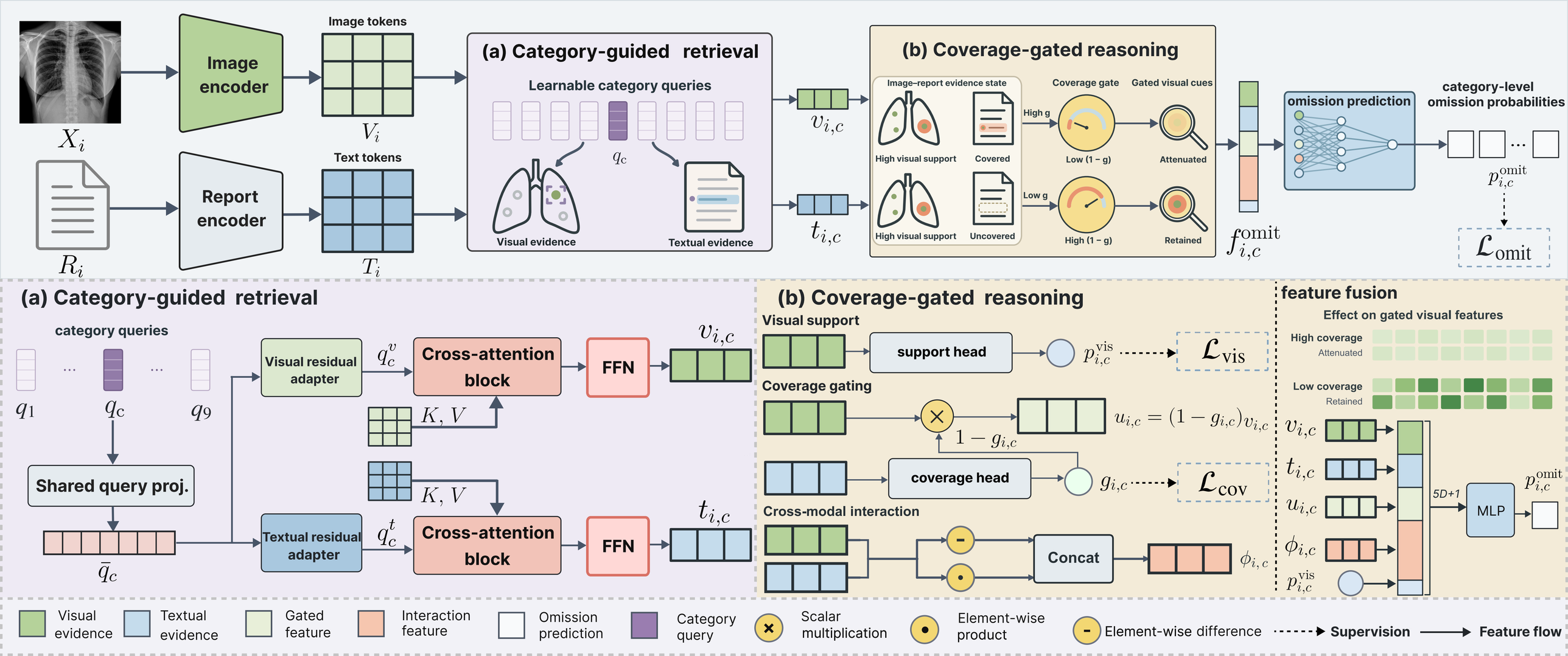

Image encoder
Image tokens
Report encoder
Text tokens
(a) Category-guided retrieval
Learnable category queries
Visual evidence
Textual evidence
(b) Coverage-gated reasoning
Image–report evidence state
High visual support
Covered
High visual support
Uncovered
Coverage gate
High g
Low (1 − g)
Low g
High (1 − g)
Gated visual cues
Attenuated
Retained
omission prediction
category-level omission probabilities
(a) Category-guided retrieval
category queries
Shared query proj.
Visual residual adapter
Textual residual adapter
Cross-attention block
Cross-attention block
FFN
FFN
(b) Coverage-gated reasoning
Visual support
support head
Coverage gating
coverage head
Cross-modal interaction
Concat
feature fusion
Effect on gated visual features
High coverage
Attenuated
Low coverage
Retained
5D+1
MLP
Visual evidence
Textual evidence
Gated feature
Interaction feature
Omission prediction
Category query
Scalar multiplication
Element-wise product
Element-wise difference
Supervision
Feature flow